

In T 110/92 the board did not dispute that the heating assembly according to claim 1 might have been a commercial success. However, such a commercial success alone, with the technically relevant examination of the claimed subject-matter leading to a negative result, could not be regarded as forming the basis for an indication of inventive step even if the board were convinced that the success derived from technical features of the heating assembly and not from other causes such as those of a commercial nature (see also T 5/91, T 219/90, T 373/94, T 109/15).

In T 478/91 too, commercial success was not regarded as indicative of inventive step. The board pointed out that it was well known that the commercial success of a product could just as easily be due to factors other than its properties, in particular more streamlined manufacture, a market monopoly, advertising campaigns or efficient selling technique (see T 270/84, T 257/91, T 712/92).

In T 1212/01 the patent related to pyrazolopyrimidinones for the treatment of impotence (Viagra). The board stated that in order to establish commercial success as an indicia of inventive step, two evidentiary steps were required. Firstly, commercial success must be demonstrated and, secondly, it had to be shown that such success resulted from the claimed invention and not from one or more other causes. Viagra had been the subject of various awards and praise in various journals. The board stated that the prizes would have been significant if awarded by persons who understood patent law. If however the prizes were awarded for the product's life-enhancing nature, or for the appellants' high standard of research, or for a high level of sales, then, for all that any of those reasons might well be prize-worthy, the prizes could have no significance in the context of inventive step. The board arrived at the conclusion that the evidence did not establish that the "prizes and praises" had resulted from the claimed inventive step.

In T 677/91 the board took the commercial success of the claimed invention into consideration and stated that it would have been wrong to ignore the practical impact that the invention had made in its own field since the priority date. For example, in several passages of a textbook, reference was made to the fact that the various advantages of the claimed invention had brought in a new era in this special field. The board concluded that it was difficult to reconcile the contents of such passages with the idea that the claimed invention was a matter of mere routine development and thus held that the invention involved an inventive step.

In T 626/96 the invention had likewise achieved great commercial success and gained widespread recognition in a number of countries. Furthermore, the success was achieved in a very short space of time, so that there seemed to be a pressing commercial need for this simple solution. Furthermore, the success was directly attributable to the structure of the product claimed and was not due to marketing techniques or advertising skills.

10.6. Market competitors

Market competitors' efforts to obtain rights of joint use constitute further secondary indicia closely related to commercial success. They may result in a positive decision on inventive step, but need not necessarily do so. In T 351/93 the board stated that the latter applied

particularly where a technical and expert assessment of the invention in the light of the prior art led, according to the problem and solution approach, to the conclusion that the existence of inventive step had to be denied.

In T 812/92 – the board found inventive step was also supported by the fact that shortly before the filing date of the contested patent, one of the patent proprietor's competitors offered a customer a technical apparatus without implementing the advantageous technical solution according to the invention.

In T 252/06 the board confirmed the presence of inventive step *inter alia* on the grounds that the patent proprietor's competitors had used the patent's teaching and had filed applications relating to it.

10.7. Simple solution

In a technical field of commercial importance to which considerable attention is directed the simplicity of a proposed solution may indicate inventive step. The difficulty of developing a simple solution without sacrificing quality may therefore indicate inventive step (T 106/84, OJ 1985, 132; T 9/86, OJ 1988, 12; see also T 229/85, OJ 1987, 237; T 44/87; T 528/89; T 73/95). This did however presuppose the absence in the prior art of anything that hinted at the proposed solution (T 712/92).

In T 29/87 the board expressed the opinion that a "simple" solution may be an indication for the presence of an inventive step if in the prior art the only processes disclosed were cumbersome, expensive, time-consuming, etc., i.e. which bore certain disadvantages and, although the problem to be solved apparently was not new and actually solved in more complex ways, the state of the art was nevertheless provided with a surprisingly "simple" solution.

In T 234/91 it was pointed out that experience in structural engineering showed that with the same or even an improved effect – in the case in point an increase in reliability – it was often much more difficult to identify and achieve a simple solution than complicated embodiments. In view of the large number of solutions suggested in the technical literature, the board came to the conclusion that the improvement achieved by the simple structural measure suggested in the contested patent had not been obvious (see also T 330/87).

In T 349/95 the subject-matter at issue related to a simple form of a simple device, a device which, however, had brought about a surprising major functional improvement. In the board's view, this surprising improvement and the fact that the prior art gave no clear hint of the solution constituted evidence of the inventiveness of the claimed solution.

A different aspect was dealt with in T 113/82 (OJ 1984, 10). In order to proceed from the known art to the invention, a series of steps needed to be taken. The board stated that this could be considered as an indicator of the presence of inventiveness, particularly in a case where the last decisive step had neither been proved to be known from the prior art nor was derivable therefrom, although this last step might at first sight seem to be a very simple one (see also T 315/87 of 13 September 1989; T 508/88; T 424/89; T 394/90).

10.8. Surprising effect – bonus effect

An effect which may be said to be unexpected can be regarded as an indication of inventive step (T 181/82, OJ 1984, 401). However, certain preconditions have to be met. In T 21/81 (OJ 1983, 15) the board considered that if, having regard to the state of the art, it would already have been obvious for a skilled person to arrive at something falling within the terms of a claim, because an advantageous effect could be expected to result from the combination of the teachings of the prior art documents, such claim lacked inventive step, irrespective of the circumstance that an extra effect (possibly unforeseen) was obtained (see T 365/86, T 350/87, T 226/88, T 1317/13). Similarly, in T 69/83 (OJ 1984, 357), the board held that where, because of an essential part of the technical problem being addressed, the state of the art obliged a skilled person to adopt a certain solution, that solution was not automatically rendered inventive by the fact that it also unexpectedly solved part of the problem. (see also T 231/97). In T 170/06 the board held that if it was obvious for the skilled person to combine prior art teachings in order to solve an essential part of the problem, the presence of even an unexpected extra effect allowing another part of the problem to be solved at the same time did not in principle imply the presence of inventive step.

Furthermore, in T 192/82 (OJ 1984, 415) the board stated that the skilled person had to be free to employ the best means already available for his purposes, although the use of means leading to some expected improvements might well be patentable in relying on an additional effect, provided this involved a choice from a multiplicity of possibilities. The lack of alternatives in this respect might therefore create a **"one-way-street"** situation leading to predictable advantages which remained obvious in spite of the existence of some unexpected **"bonus" effect** (see also T 766/92, T 1936/13). The board also pointed out in T 506/92 that an additional effect achieved inevitably by the skilled person on the basis of an obvious measure without any effort on his part simply represented a bonus under EPO case law which could not substantiate inventive step, even as a surprising effect (see also T 431/93, T 681/94, T 985/98, T 794/01, T 688/13, T 179/18). In T 848/94 the solution of the existing technical problem required a combination of measures that was not suggested by the prior art in such a manner that it would have been adopted by the person skilled in the art (see also T 716/07). Therefore, the person skilled in the art was not in a "one-way-situation".

The board in T 936/96 held that, once a realistic technical problem had been defined and once it had been established that a particular solution to such a problem would have been envisaged by a skilled person in the light of the relevant state of the art, that solution could not be said to involve an inventive step, and this assessment was not altered by the fact that the claimed invention inherently also solved further technical problems. In the case in point the claimed surprising effect could not be regarded as an indication of the presence of an inventive step.

In T 227/89 the board stated that in determining which effect was crucial and which was merely accidental (the so-called "bonus effect"), a realistic approach had to be taken, considering the relative technical and practical importance of those effects in the circumstances of a given case (see also T 732/89, T 729/90, T 1147/16). When assessing

chemical substances for inventive step, it is often their surprising properties that are considered (see in this connection T.20/83, OJ 1983, 419).

According to the case law of the boards of appeal, an improvement is not a prerequisite for inventive step (see also in this chapter I.D.4.5 "Alternative solution to a known problem"). In T.154/87 it was pointed out that the achievement of a surprising effect was no precondition for the existence of inventive step. All that was necessary was to ascertain that the respective subject-matter could not be derived by the skilled person in an obvious manner from the available prior art (T.426/92, T.164/94, T.960/95, T.524/97, T.888/08).

In T.551/89 the board stated that an effect which was to be expected as the result of an obvious measure could not contribute to recognition of the required inventive step, even if the scale of this effect was surprising to the skilled person. In this case an effect whose scale surpassed the skilled person's hopes merely represented a bonus effect following inevitably from the use of an obvious measure and obtained by the skilled person without any inventive effort on his part (T.506/92, T.882/94).

In T.240/93 the application related to an apparatus for the surgical treatment of tissues by hyperthermia, equipped with heat protection means. The application was refused by the examining division, which considered the short treatment duration of one hour and further advantages resulting from the use of cooling means to be extra (bonus) effects. The board, however, stated that in the case in point the objective problem underlying the invention was to provide an apparatus for the effective therapeutic treatment of benign prostate hyperplasia in a short period of time. In view of the many considerable practical advantages of a single one-hour hyperthermia session for a patient, such a short treatment duration could not be dismissed as a mere "bonus" effect, but was crucial to the invention and the basis of the objective problem.

In T.2015/20 the application stated that it had been surprisingly found that for treatment of respiratory disorders, in particular asthma and COPD, acclidinium was most effective upon administration by inhalation in a dosage of about 400 µg metered nominal dose. D1 described the combination of a M3 muscarinic receptor antagonist such as acclidinium bromide with a PDE4 inhibitor, and described 1-2 formulations comprising 100 µg acclidinium bromide. It did not provide any suggestion towards an optimized dose of 400 µg. D2 presented a short summary of a trial in which patients suffering from COPD were administered a single dose of 100, 300 or 900 µg of acclidinium bromide, but which the board concluded seemed to teach away from an optimised dose of 400 µg for treatment of a chronic disease such as asthma. The board found that the defined subject-matter of claim 1 was not the obvious result of routine experimentation, but rather represented the unexpected outcome of a study and concluded that the subject-matter of claim 1 involved an inventive step.

E. The requirement of industrial application under Article 57 EPC

1. Notion of "industrial application"	302
1.1. Invention and industrial application	302
1.2. Any kind of industry	303
1.2.1 Methods applied in the private and personal sphere	304
1.2.2 Cosmetic and beauty salons	305
2. Reproducibility and sufficiency of disclosure	305
3. Indication of a profitable use of the invention in industry	306

1. Notion of "industrial application"

Art. 57 EPC provides that: "An invention shall be considered as susceptible of industrial application if it can be made or used in any kind of industry, including agriculture". It thus defines the third criterion for patent grant referred to in Art. 52(1) EPC, namely "industrial application", making it clear that the possibility of making or using the invention in any kind of industry is sufficient (T 144/83) and that agriculture too is a kind of industry for the purposes of the EPC (T 116/85, OJ 1989, 13). Both products and processes (T 208/84) are susceptible of industrial application. Claims directed to substances or compositions for use in methods for treatment of the human or animal body by therapy are directed to inventions which are susceptible of industrial application (G 5/83, OJ 1985, 64).

In T 80/96 (OJ 2000, 50), the board found that the use of a substance to make a new pharmaceutical product without delimitation to an indication did not contravene the requirements of Art. 57 EPC 1973 in conjunction with Art. 52(1) EPC 1973.

Regarding the amendments to Art. 52(1) EPC in the course of the revision of the EPC, see chapter I.A.1. "Patent protection for technical inventions".

1.1. Invention and industrial application

The requirement of industrial application must be satisfied in addition to the other patentability criteria. Art. 57 EPC excludes only a few inventions from patentability that are not already excluded by Art. 52(2) EPC, or by Art. 53 EPC. In particular the work of medical practitioners, previously excluded from patentability under Art. 52(4) EPC 1973 as lacking industrial application, is now covered in Art. 53(c) EPC. Industrial application is also lacking if the product or process is contrary to the laws of physics (T 541/96), such as for example a perpetual motion machine (Guidelines G-III, 3 – March 2022 version).

While the technical character of an invention and its susceptibility of industrial application are two different patentability requirements under the EPC, they are often discussed together.

In T 953/94 the board stated that, as the Convention showed (Art. 52(1) EPC 1973), the requirement (defined in Art. 52(2) and (3) EPC 1973) of claimed subject-matter being an